

Hong Yan Jun | CAD Portfolio

Mechanical Design

Project 1 | Sound Tracking and Localization Sentry

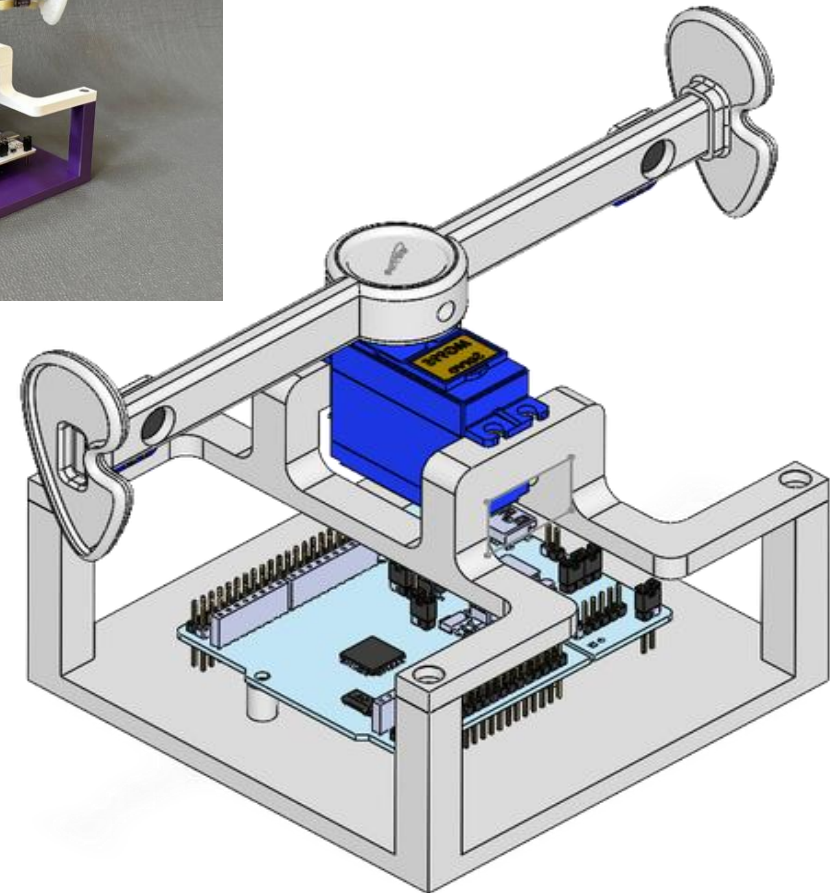
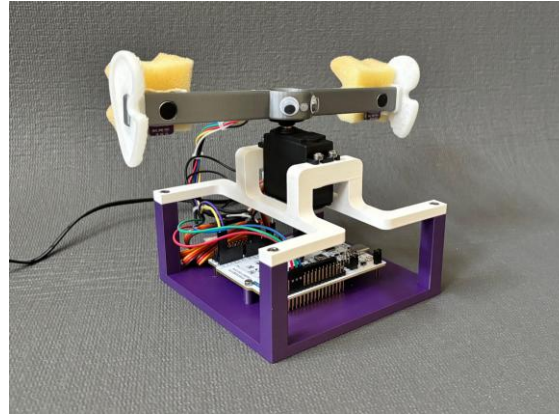
Project Overview

- **Objective:** To build a system that mimics the human hearing mechanism.
- **Description:** A real-time sound tracking system that turns to the direction of the source of sound.
- **Components:** Microphones, Servo motor, Microcontroller, Bar, Base.

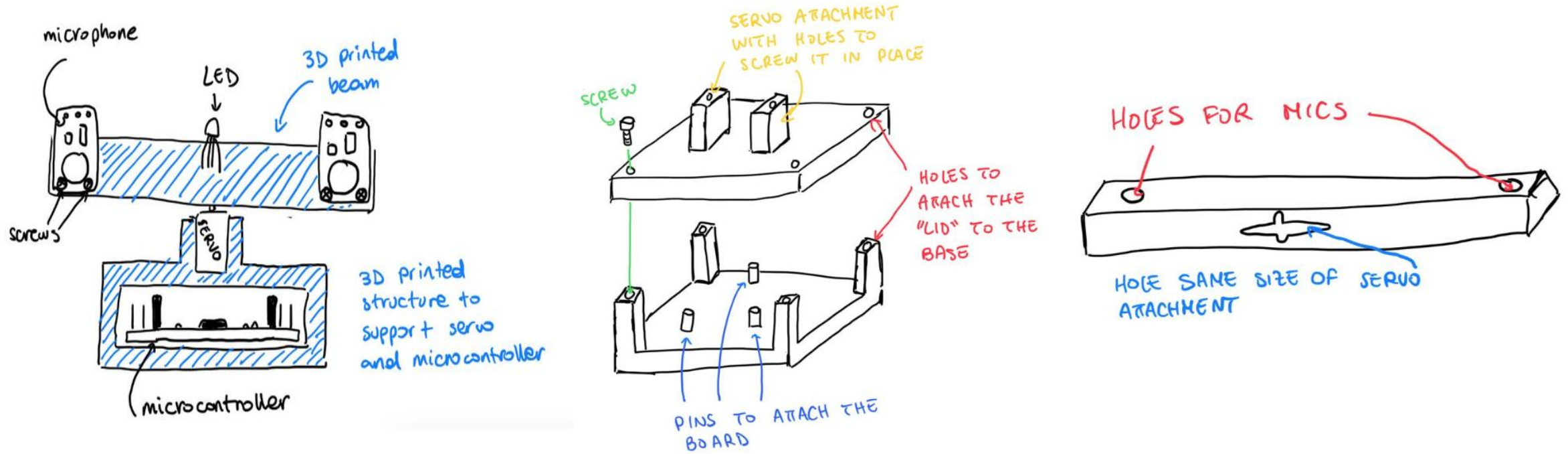
Project Details

CAD Software	Solidworks 2026
3D Printing Material	Polylactic acid (PLA)
Number of Parts	10
Timeline	June – July 2026
Methods Involved	Static FEA

Final Design



Project 1 | Initial Concept Sketches



The very first brainstorm sketches of a simple functional platform to host the microelectronics components.

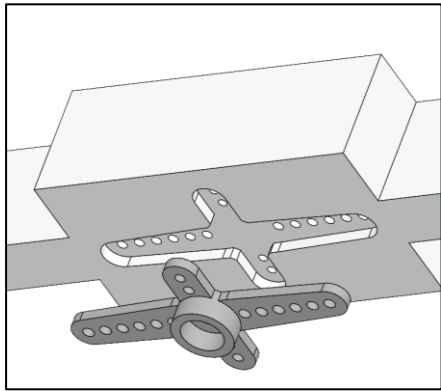
Key Design Requirements:

- 15cm distance between the 2 microphones (To mimic the distance of human ears).
- The weight of the platform must be able to support the bar and microphones.

Project 1 | Initial CAD Model and Design Choices

Microphone mounting apertures tolerance-fit to component outer diameter for press-fit stability without secondary fasteners.

Tunnels for the wires of servo motor to pass through so that it could fit between the walls.

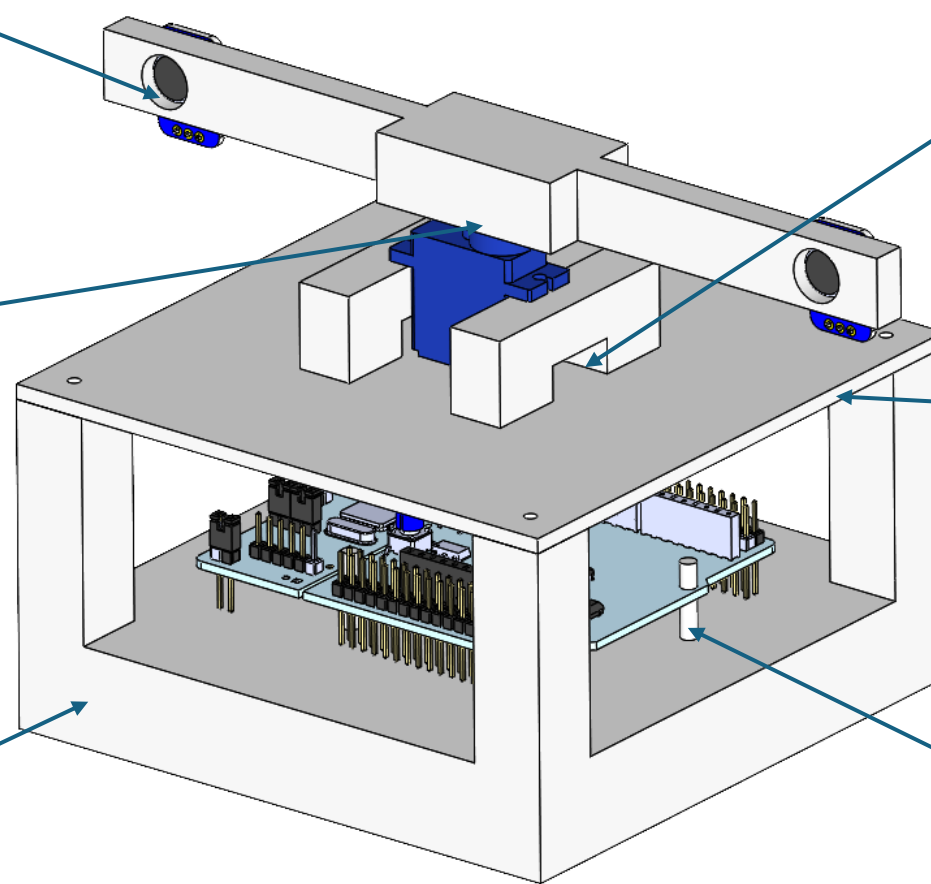


The bottom of the bar was designed to fit the servo motor head nicely.

The top was made thin initially so the screws could pass through easily.

Initial base geometry weighted to provide a low center of gravity and resist dynamic tip-over moments during rapid servo slewing.

Poles for fixing the position of the microcontroller on the platform.



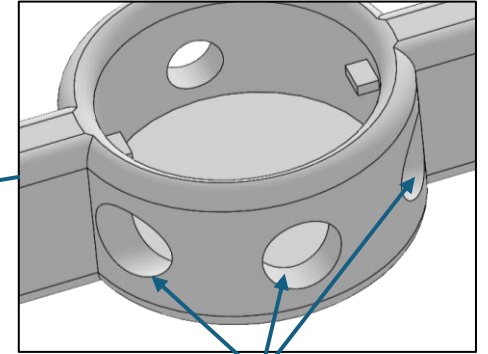
Project 1 | Final CAD Model and Key Design Choices Changes

A bigger servo motor was chosen at the end. Hence, the top was redesigned accordingly.

Top part was made thicker to support the bigger motor and screw holes were modified so they could bite into the plate.

A lid was added so that wires could be hidden inside the bar.

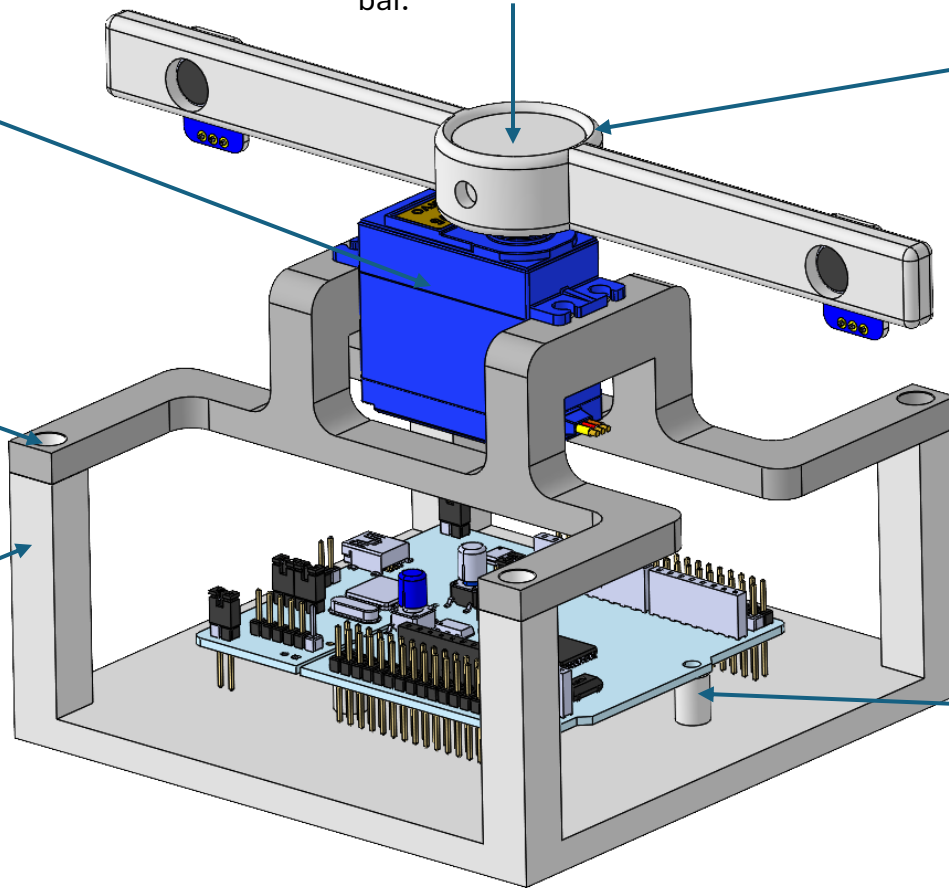
Since the weight requirement is fulfilled, the top and bottom were redesigned to save PLA printing filaments and improve platform aesthetic.



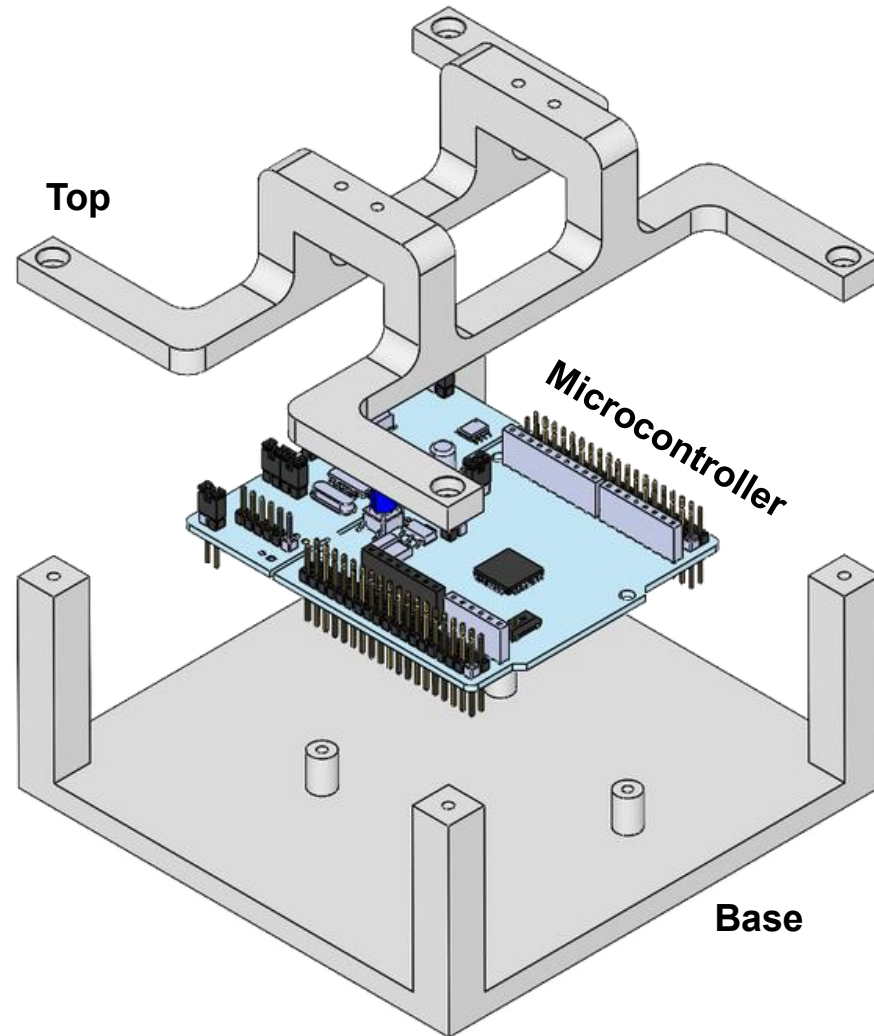
These holes were added to separate the wires going to the left and right microphones for cleaner cable management.

Additional screw holes were added to firmly bind the microcontroller to the platform.

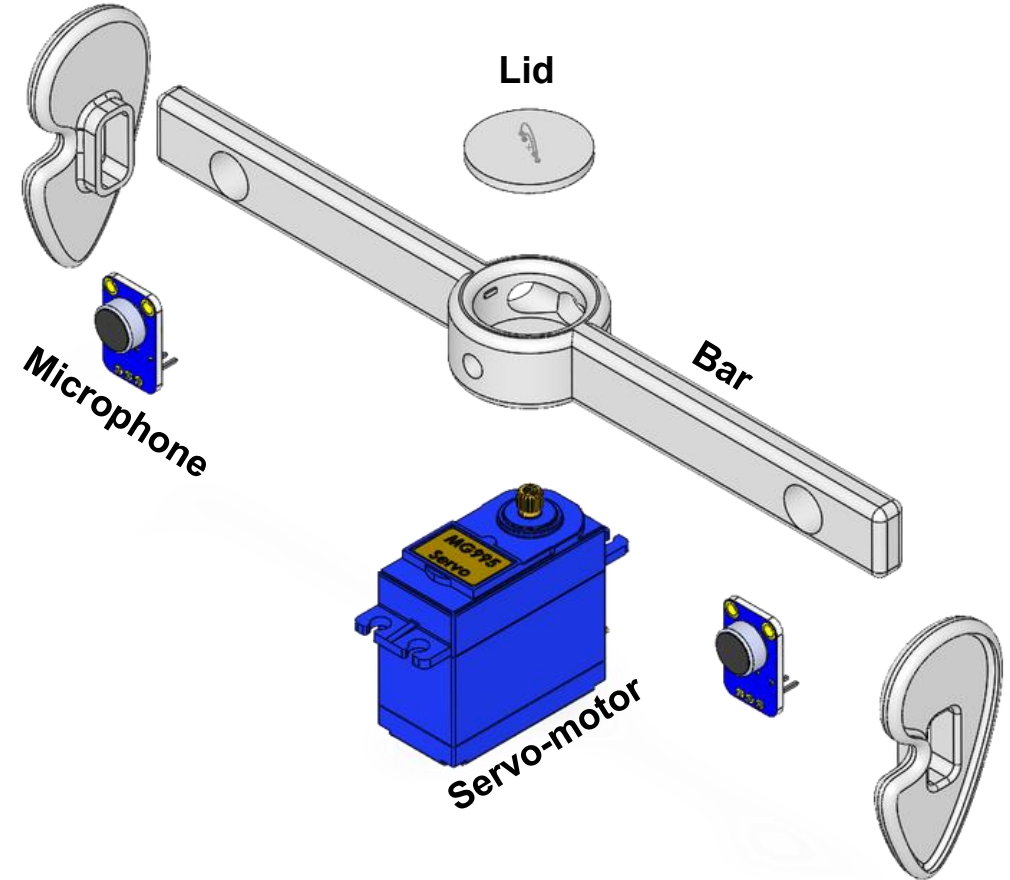
Applied edge fillets to eliminate stress concentrations and reduce visual seam harshness.



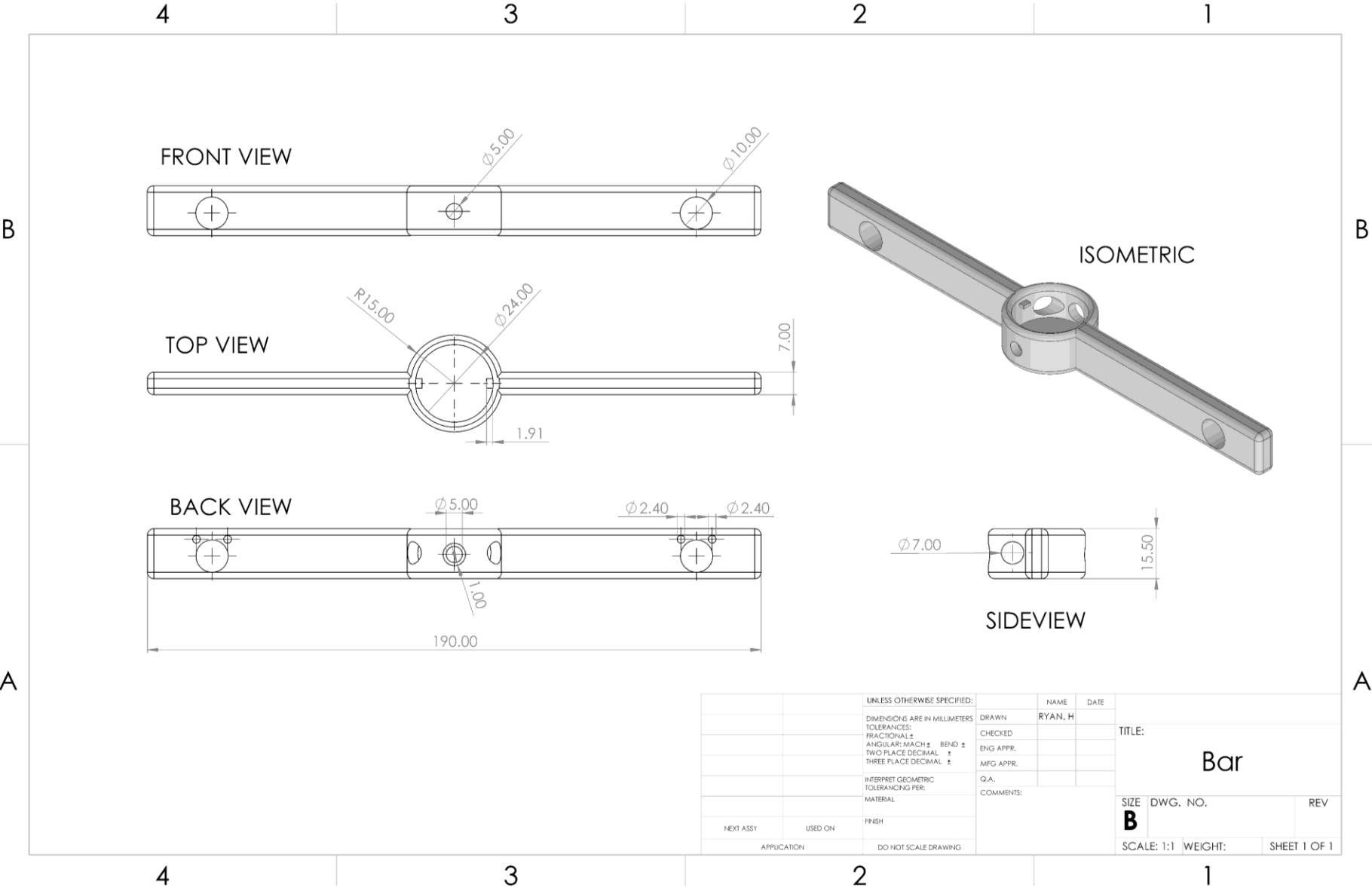
Project 1 | Exploded View of Individual Components



Ear - Acoustic Baffle



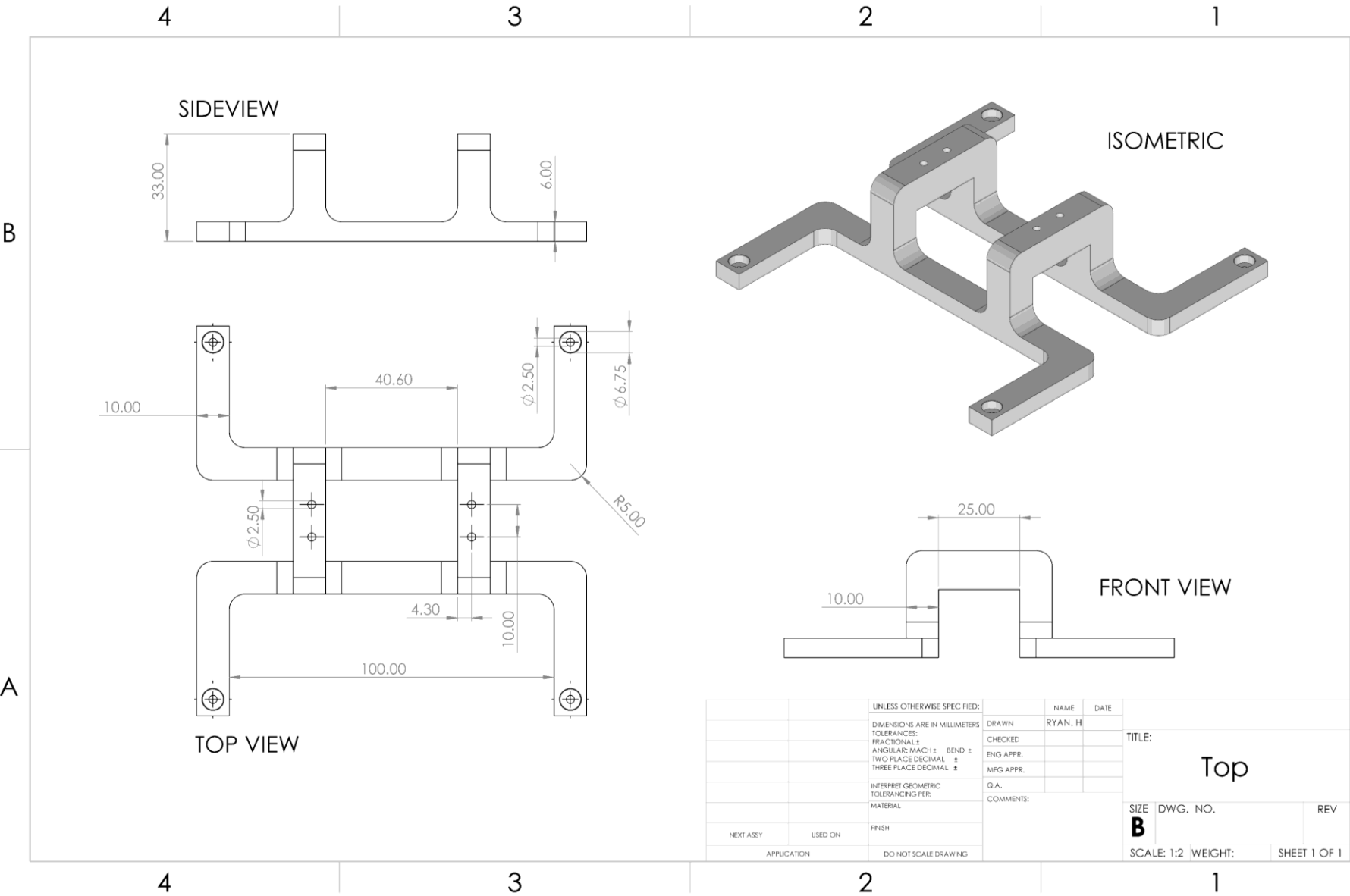
Project 1 | Technical Drawing of Microphone Holder Bar



	Details
CAD Software	Solidworks
Material	PLA
Number of Components	1
Timeline	June – July 2026

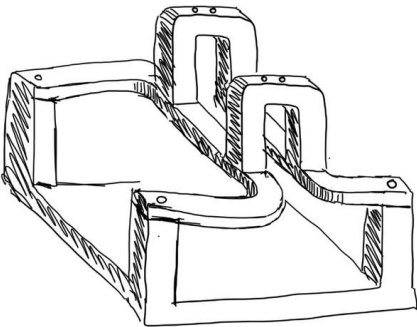
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Project 1 | Technical Drawing of Servo Holder Top



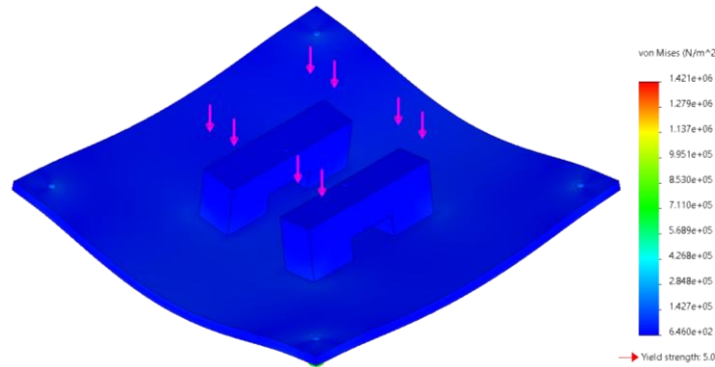
	Details
CAD Software	Solidworks
Material	PLA
Number of Components	1
Timeline	June – July 2026

Final design sketch:



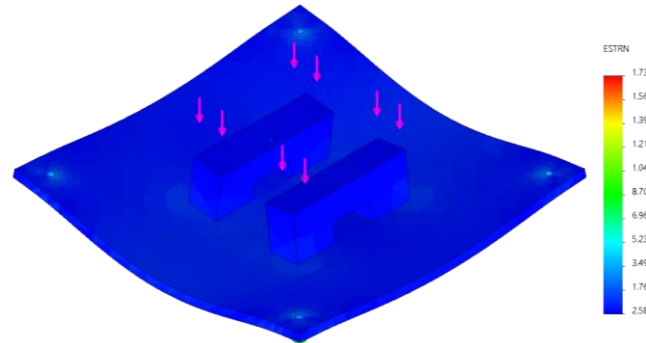
Project 1 | Static FEA Analysis of Initial Model

Stress



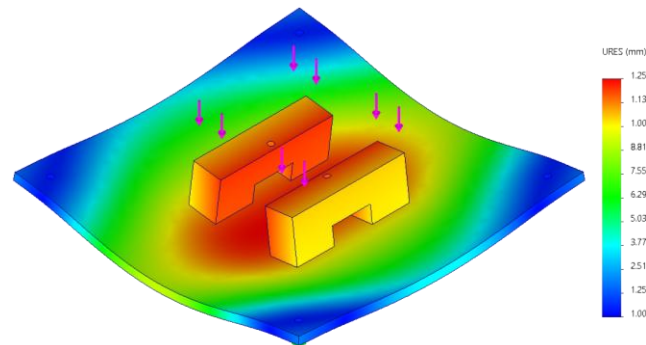
- Maximum stress of **1.42MPa** far below yield strength of **50MPa**.
- Safe factor **n = 35**

Strain



- Maximum strain of **0.0001738**, which is negligible.

Displacement



- Maximum displacement of **0.013mm**, which is negligible.

Details

Material	PLA
Elastic Modulus	3500 N/mm ²
Poisson's Ratio	0.33 N/A
Mass Density	1240 kg/m ³
Load applied	0.93 N
Printing Time	1h1m
Filament Used	41.96g

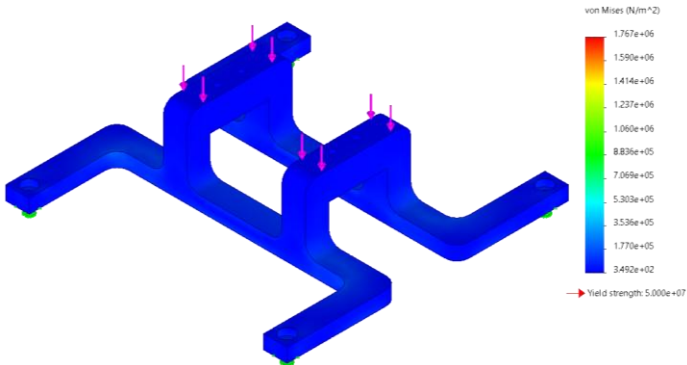
Initial static analysis revealed a massive **Factor of Safety of 35**. While structurally sound, the design was highly over-engineered, making it **inefficient in both material consumption and production time**. This justified a complete geometric redesign to optimize for manufacturing

Project 1 | Static FEA Analysis of Final Model

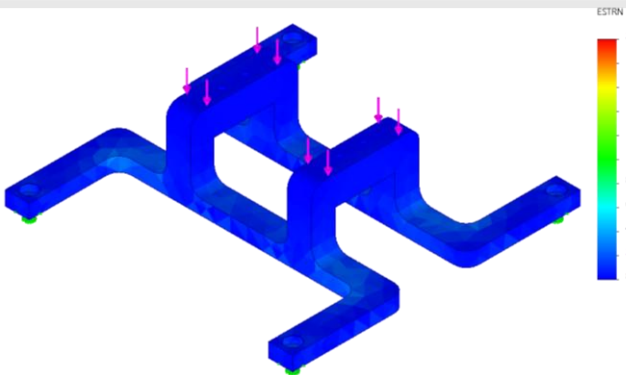
Stress

Strain

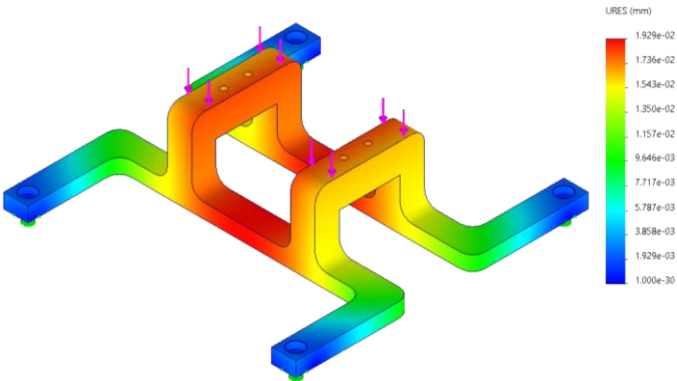
Displacement



- Maximum stress of **1.767MPa** far below yield strength of **50MPa**.
- Safe factor **n = 28**



- Maximum strain of **0.0002218**, which is negligible.



- Maximum displacement of **0.019mm**, which is negligible.

	Details
Material	PLA
Elastic Modulus	3500 N/mm²
Poisson's Ratio	0.33 N/A
Mass Density	1240 kg/m³
Load applied	0.93 N
Printing Time	50m44s
Filament Used	24.53g

By accepting a negligible increase in stress (from **1.42 MPa to 1.76 MPa**), the redesigned bridge structure achieved a **41.5% reduction in material usage** and shaved roughly **10 minutes off the production cycle time per unit**, while simultaneously improving the overall aesthetic.

Project 2 | V8 Engine Assembly

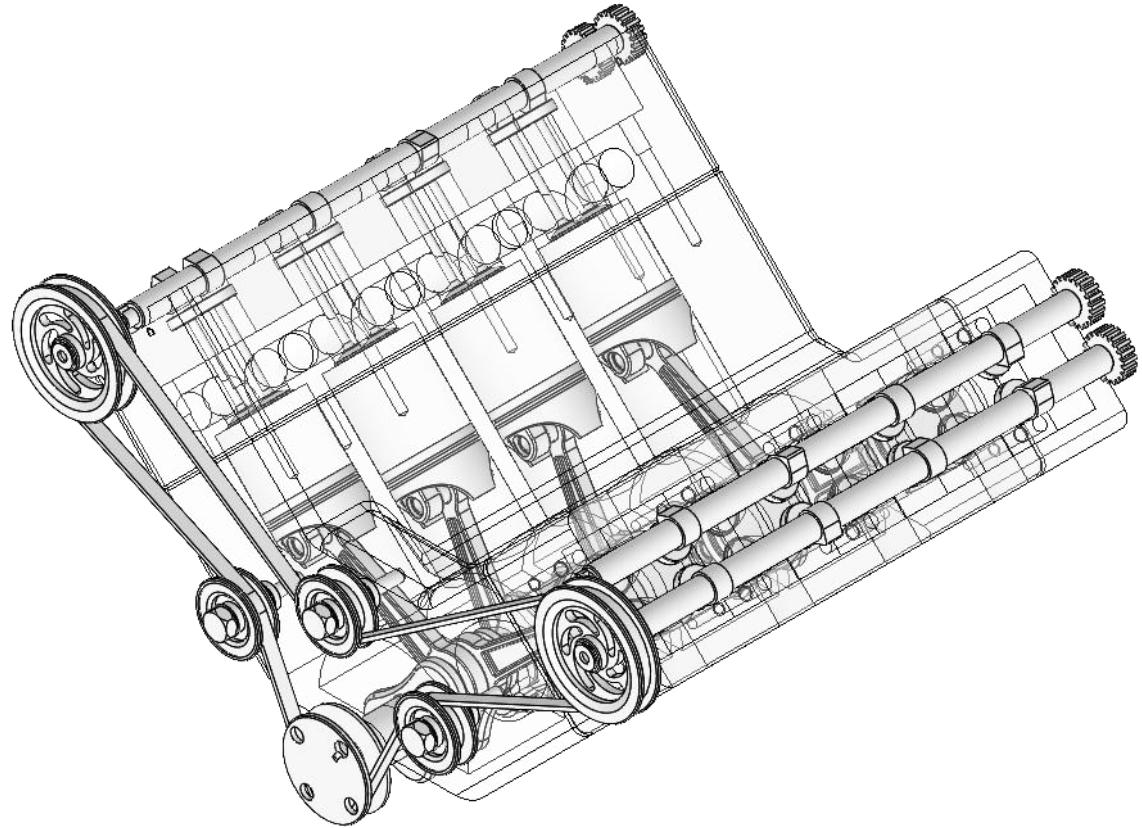
Project Overview

- **Objective:** To build a functional simplified V8 engine.
- **Description:** Naturally aspirated 90-degree V8 internal combustion engine assembly (2.46L displacement), featuring an integrated valvetrain, cross-plane crankshaft, and multi-body kinematic timing drive.
- **Components:** Cylinder Block, Crankshaft, Pistons, Cylinder Heads, Camshafts.

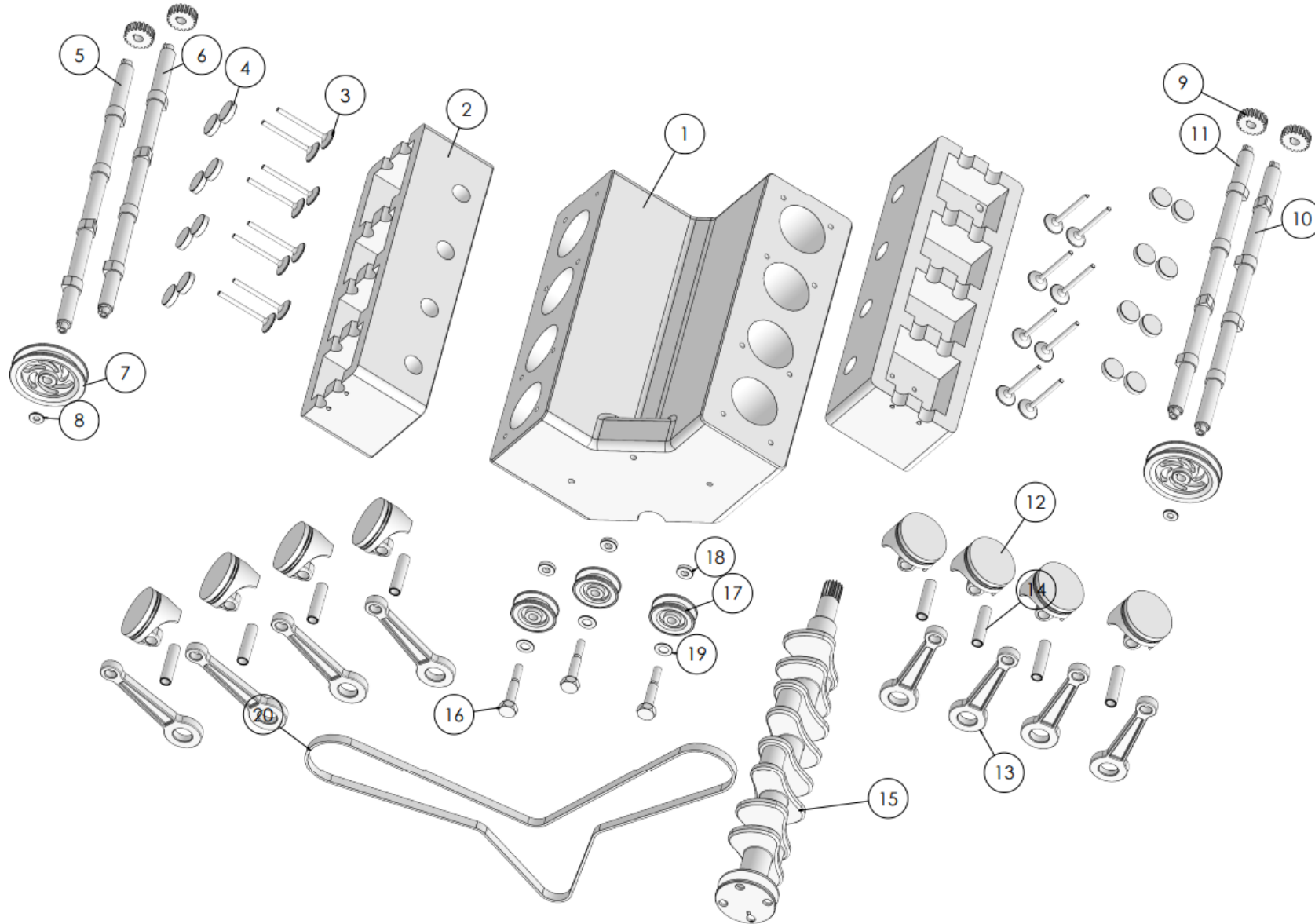
Project Details

CAD Software	Solidworks 2026
Number of Parts	89
Timeline	August 2026
Methods Involved	GD&T

Final Design

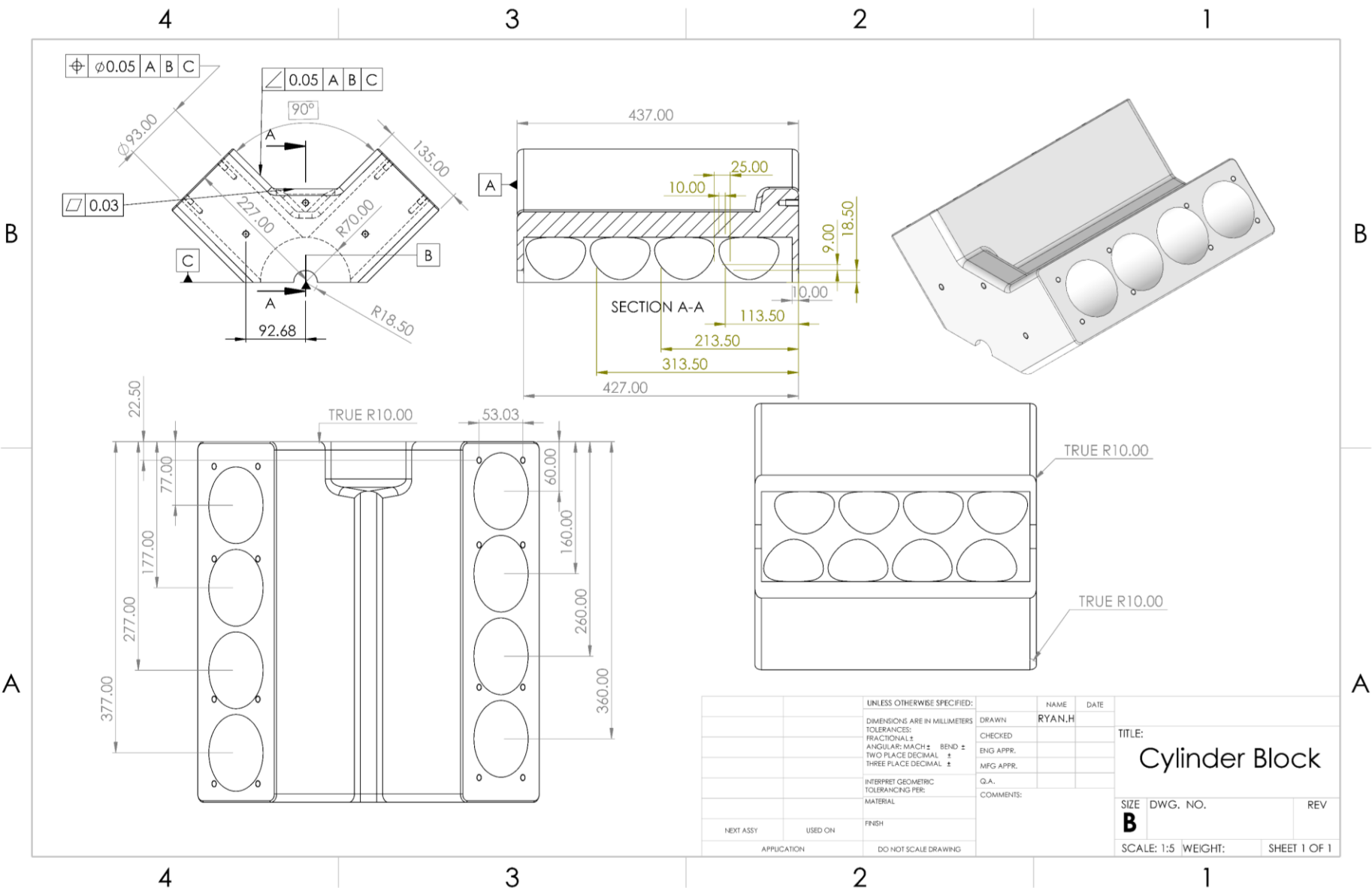


Project 2 | Exploded view of individual components



ITEM NO.	PART NUMBER	QTY.
1	Cylinder Block	1
2	Cylinder Head	2
3	Valve	16
4	Valve Washer	16
5	Camshaft - Exhaust 1-2-3-4	1
6	Camshaft - Intake 1-2-3-4	1
7	Belt Pulley	2
8	Pulley Washer	2
9	Gear	4
10	Camshaft - Exhaust 5-6-7-8	1
11	Camshaft - Intake 5-6-7-8	1
12	Piston	8
13	Connecting Rod	8
14	Piston Pin	8
15	Crankshaft	1
16	Custom Screw M10	3
17	Idler Pulley	3
18	Spacer Washer	3
19	Idler Pulley Washer	3
20	Timing Belt	1

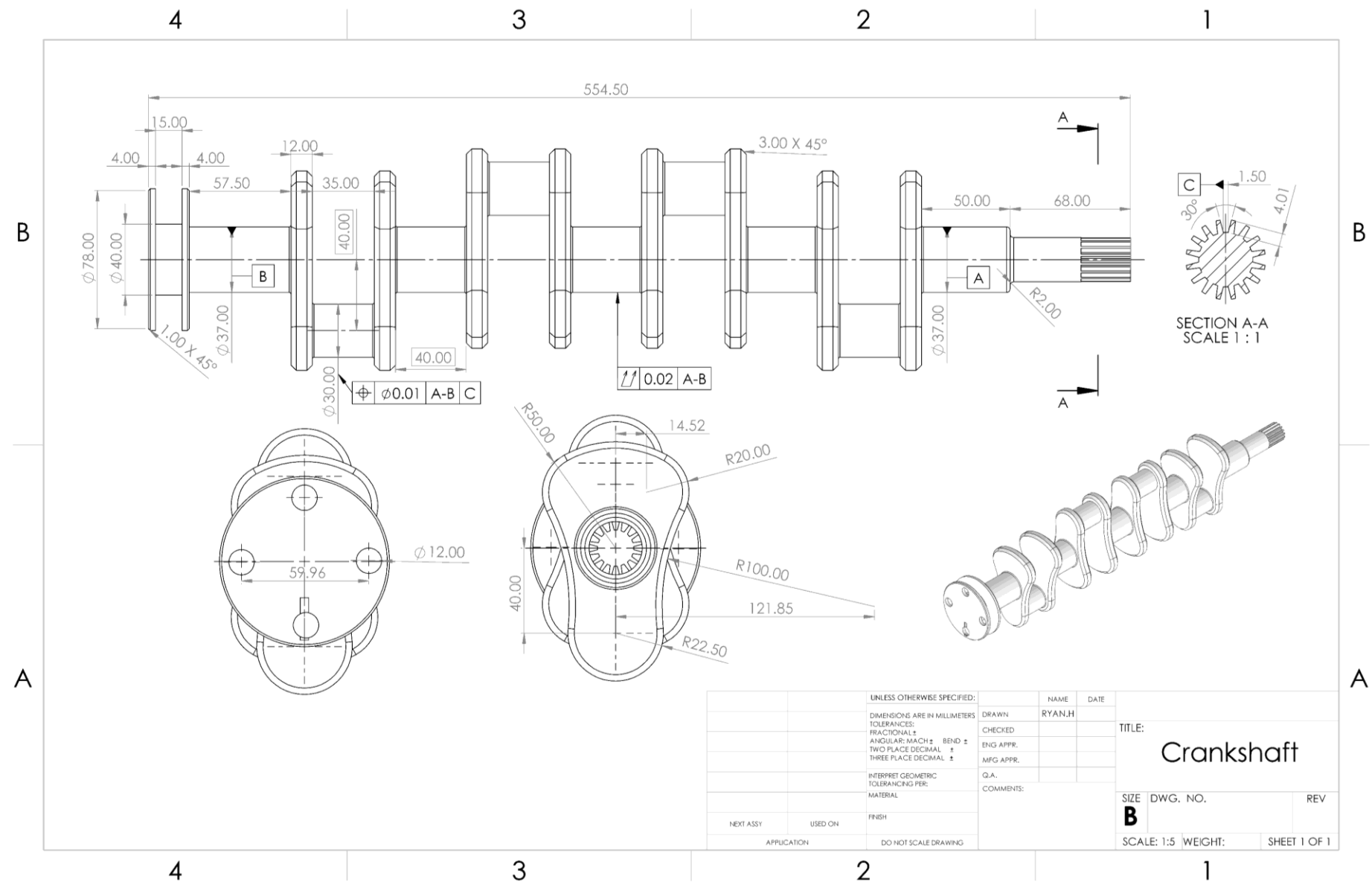
Project 2 | Technical drawing of Cylinder Block



	Details
CAD Software	Solidworks
Number of Components	1
Timeline	August 2026

- Engine Layout: 90-degree V8.
- The block features a classic, balanced 90-degree V-angle between the cylinder banks
- Displacement: 2,463 cc (2.5 Liters / 150 CID).

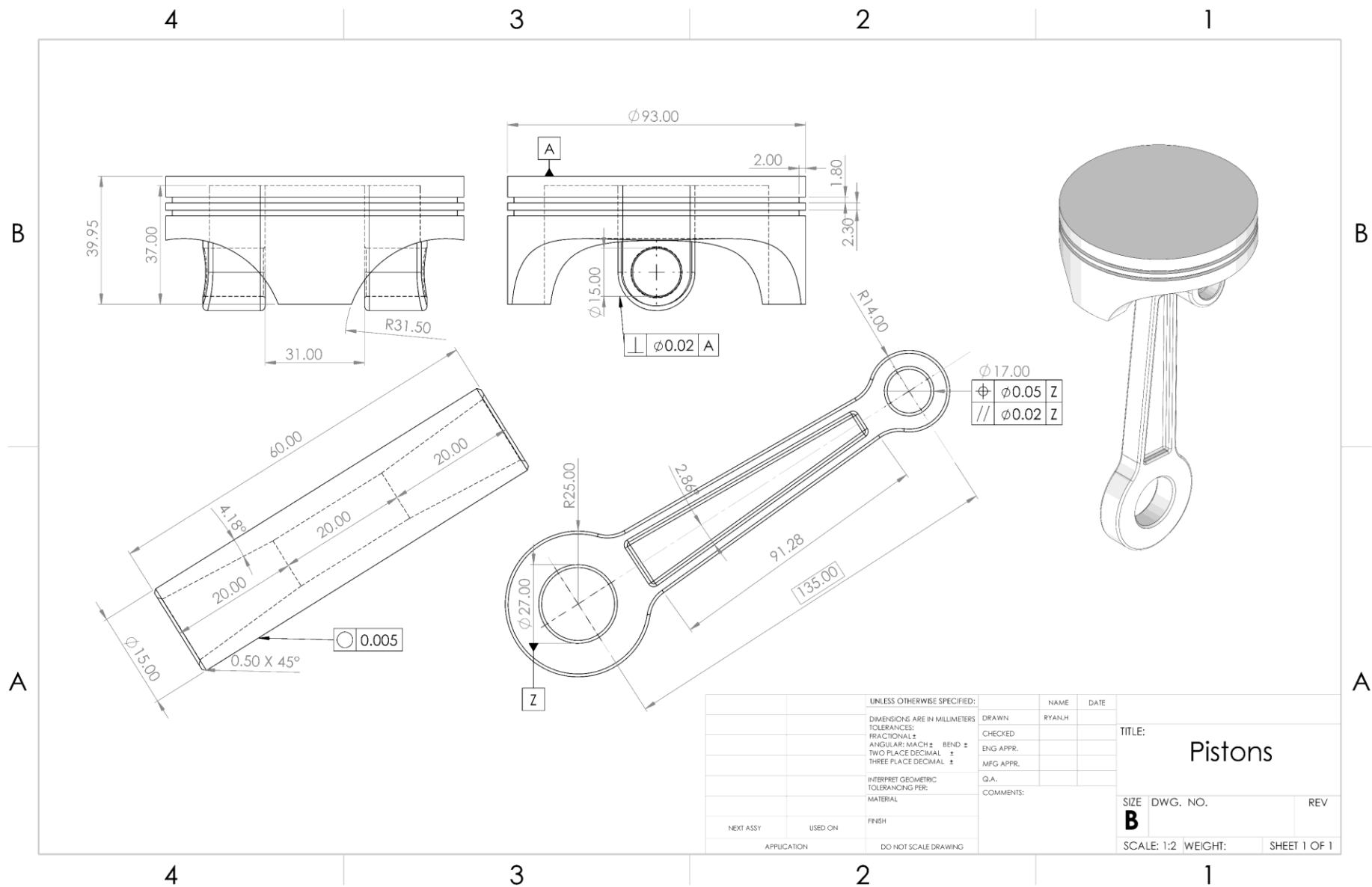
Project 2 | Technical drawing of Crankshaft



	Details
CAD Software	Solidworks
Number of Components	1
Timeline	August 2026

UNLESS OTHERWISE SPECIFIED:		NAME	DATE	TITLE: Crankshaft	
DIMENSIONS ARE IN MILLIMETERS		DRAWN	RYAN.H		
TOLERANCES:		CHECKED			
FRACTIONAL ±		ENG APPR.			
ANGULAR: MACH ± BEND ±		MFG APPR.		SIZE DWG. NO. REV	
TWO PLACE DECIMAL ±		G.A.			
THREE PLACE DECIMAL ±		COMMENTS:		SCALE: 1:5 WEIGHT: SHEET 1 OF 1	
INTERPRET GEOMETRIC TOLERANCING PER:					
MATERIAL					
NEXT ASSY	USED ON	FINISH			
APPLICATION	DO NOT SCALE DRAWING				

Project 2 | Technical drawing of Pistons & Connecting Rods



	Details
CAD Software	Solidworks
Number of Components	3
Timeline	August 2026

UNLESS OTHERWISE SPECIFIED:		NAME	DATE
DIMENSIONS ARE IN MILLIMETERS		RYAN.H	
TOLERANCES:			
FRACTIONAL ±		CHECKED	
ANGULAR: MACH ±	BEND ±	ENG APPR.	
TWO PLACE DECIMAL ±		MFG APPR.	
THREE PLACE DECIMAL ±		Q.A.	
INTERPRET GEOMETRIC TOLERANCING PER:		COMMENTS:	
MATERIAL			
FINISH			
NEXT ASSY	USED ON		
APPLICATION	DO NOT SCALE DRAWING		
TITLE: Pistons		SIZE B	DWG. NO.
SCALE: 1:2		WEIGHT:	REV
			SHEET 1 OF 1

Project 3 | Modular Kinematic Marble Run

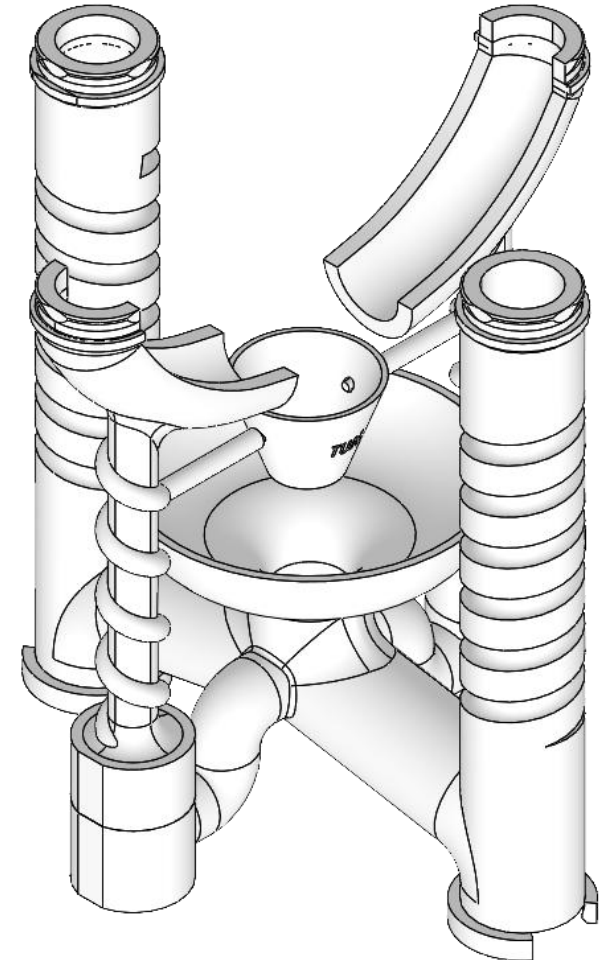
Project Overview

- **Objective:** To build a marble run module with additive manufacturing.
- **Description:** Marble run module with top bottom inputs and left right outputs integrated with self pivoting tipping bucket mechanism in-between.
- **Components:** Base bowl, Slide pillars and Bucket.

Project Details

CAD Software	Solidworks 2026
3D Printing Material	Polylactic acid (PLA)
Number of Parts	3
Timeline	September 2026
Methods Involved	DFAM

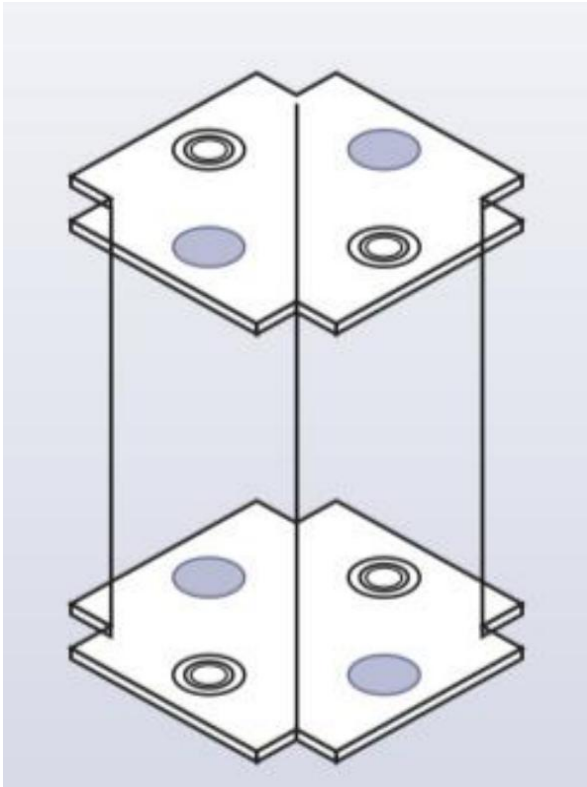
Final Design



Project 3 | Modular Kinematic Marble Run

Requirements:

- 2 inputs and 2 outputs
- Any system that triggers a mechanical process with a marble.

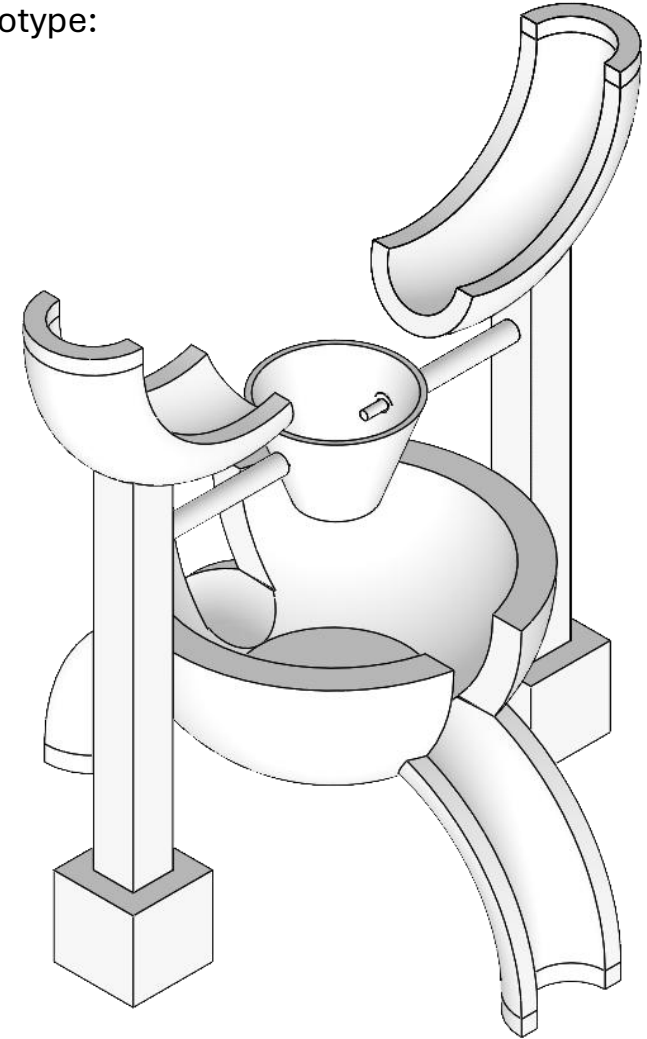


Idea:

- Tipping bucket from a waterpark.
- Physics: Center of mass below axle rod.

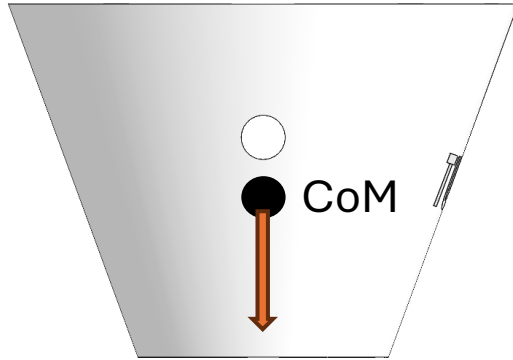


First Prototype:



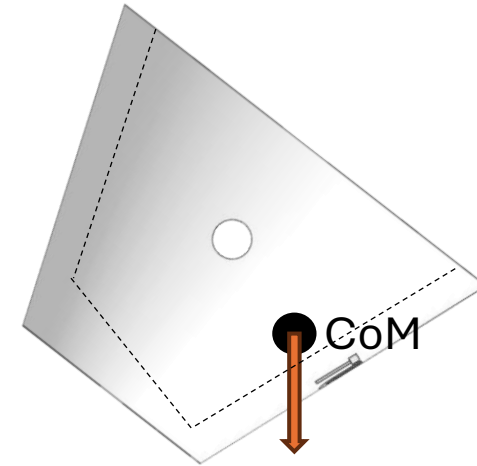
Project 3 | The Tipping Bucket Mechanism

Empty & Stable:



The pivot (14.00mm) is positioned well above the geometric midpoint of the 22.50mm tall bucket, so most of the mass hangs below the axle, which creates the restoring counterweight.

Loaded & Tipping:



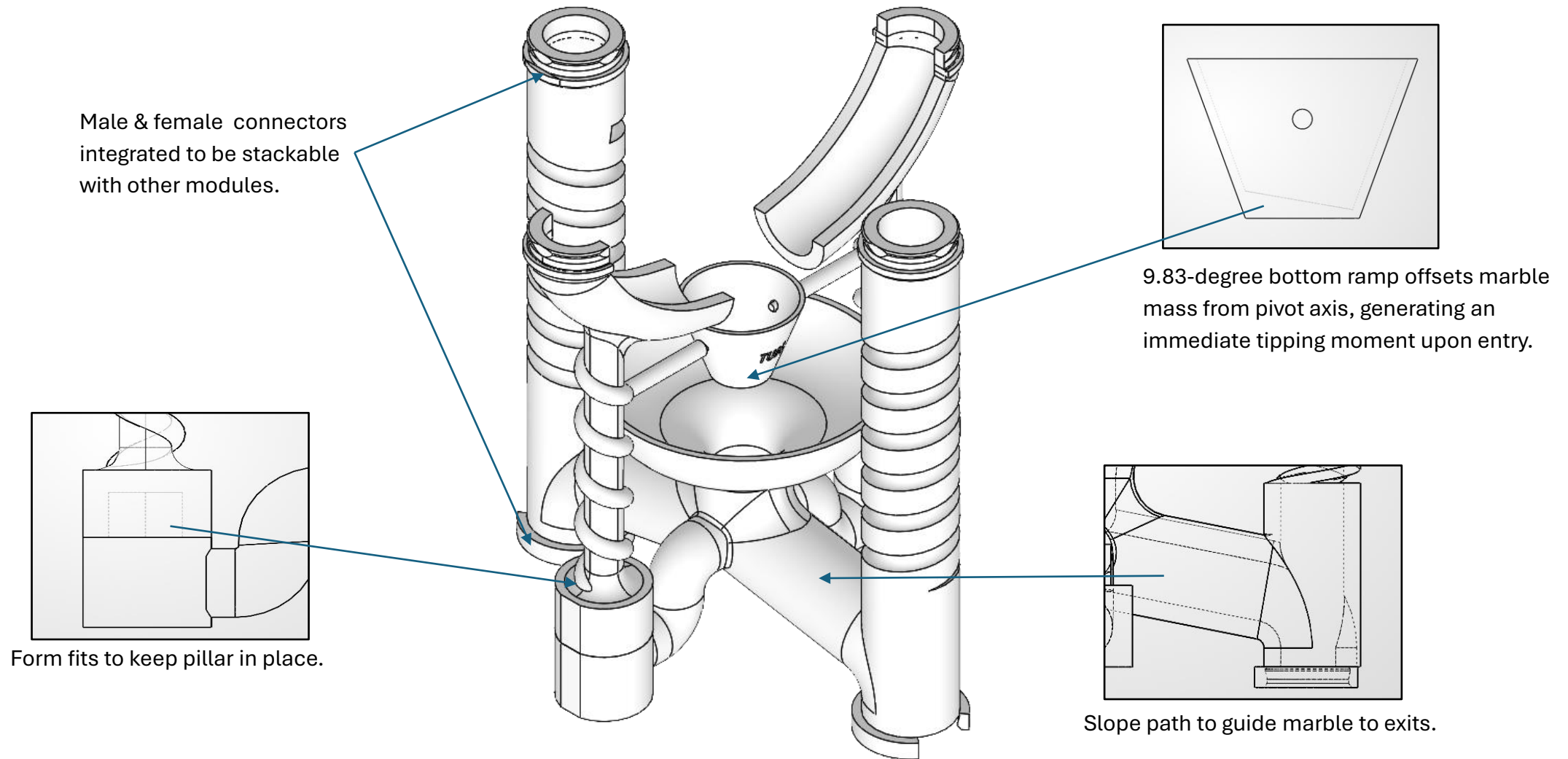
The 9.83° incline floor at the bottom not only allows the marble to roll out smoothly, but it forces the incoming 15-16mm marble to rest off-centre relative to the pivot axis.

Governing Dynamics:

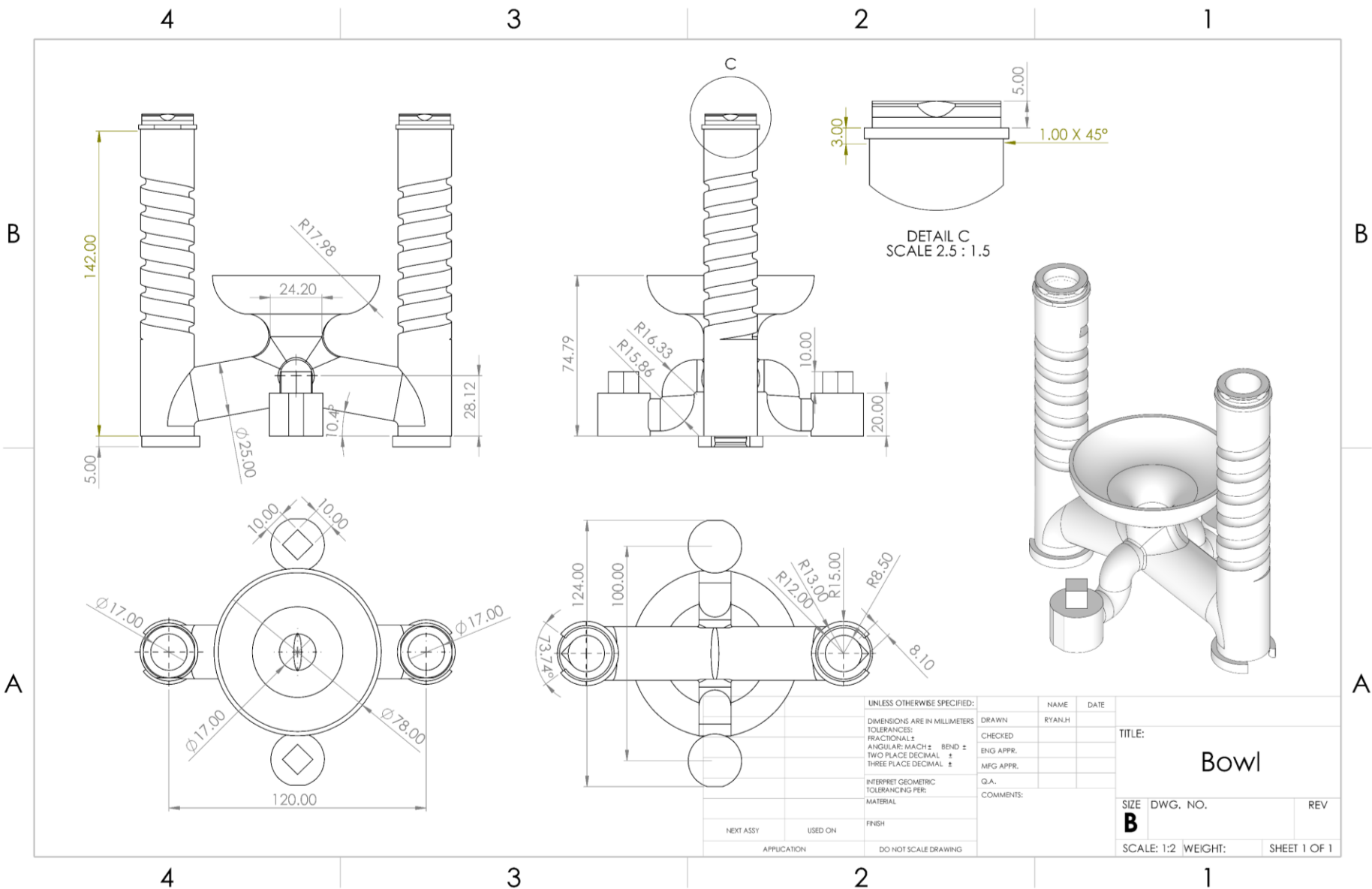
$$\sum \tau_{pivot} = (m_{marble} \cdot g \cdot x_{marble}) - (m_{bucket} \cdot g \cdot x_{bucket})$$

- m_{marble} / m_{bucket} : Mass of the 16 mm marble vs. mass of the PLA bucket.
- x_{marble} : The horizontal distance from the pivot to the marble's CoM (driven by the 9.83° floor slope).
- x_{bucket} : The dynamic horizontal shift of the bucket's CoM as it begins to rotate.
- **Design Conclusion:** The bucket tips when $\sum \tau_{pivot} > 0$, and the 14.00 mm pivot height was mathematically tuned to ensure the bucket immediately rights itself once m_{marble} exits the system.

Project 3 | Final CAD Model and Key Design Choices Changes



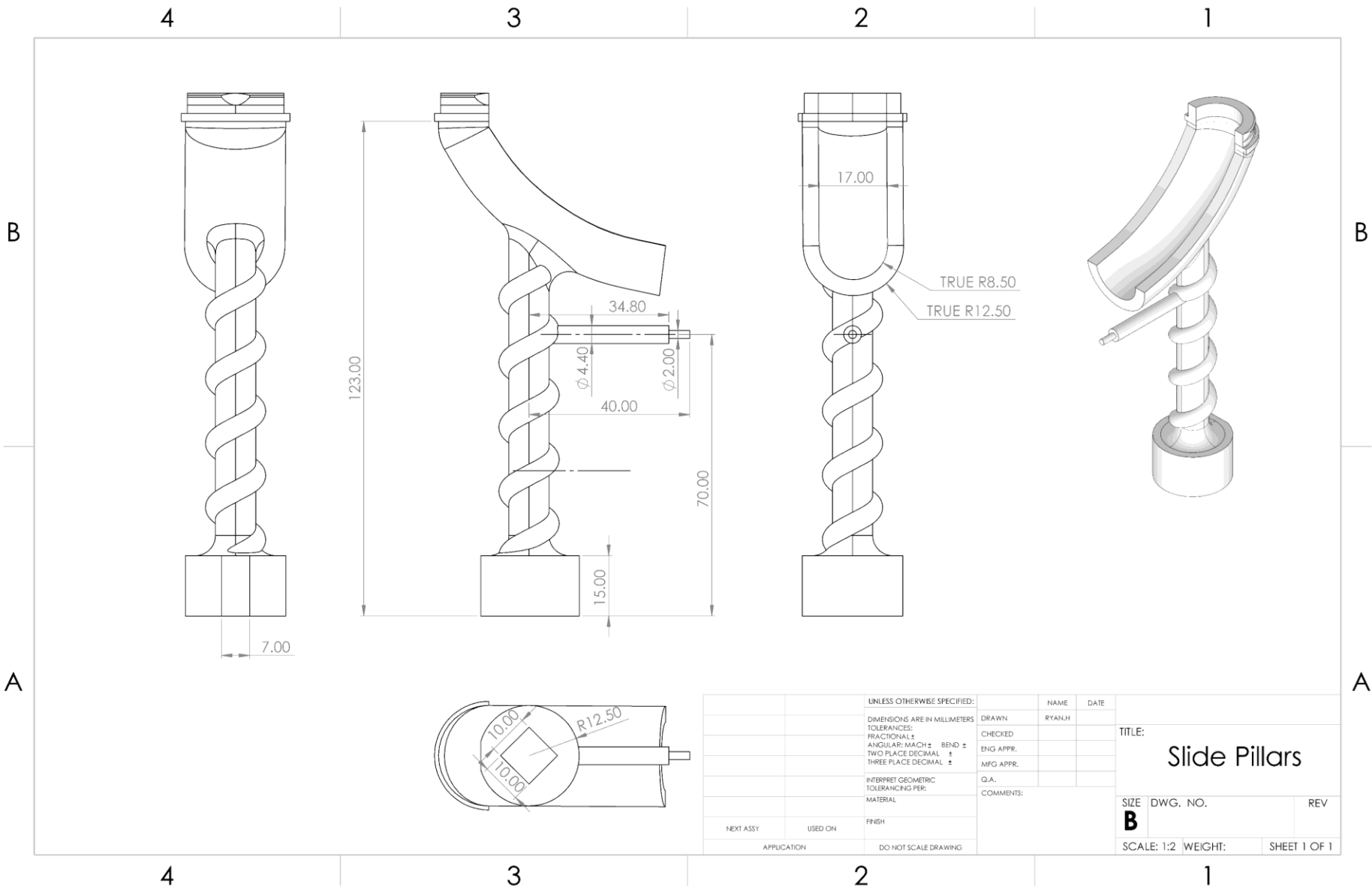
Project 3 | Technical drawing of Bowl Base



	Details
CAD Software	Solidworks
Number of Components	1
Timeline	September 2026

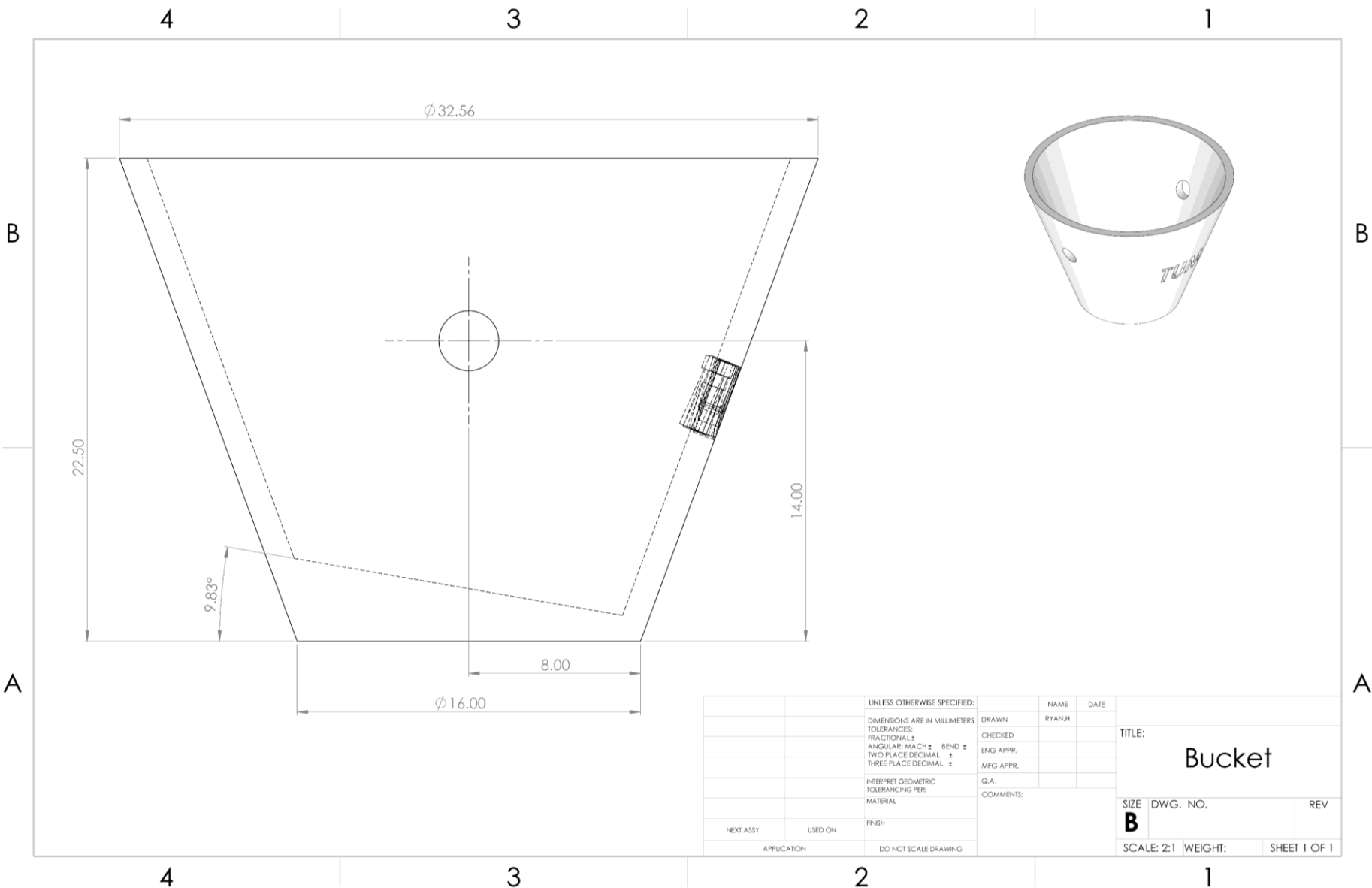
UNLESS OTHERWISE SPECIFIED:		NAME	DATE
DIMENSIONS ARE IN MILLIMETERS		RYAN.H	
TOLERANCES:			
FRACTIONAL:			
ANGULAR: MACH ± BEND ±			
TWO PLACE DECIMAL ±			
THREE PLACE DECIMAL ±			
INTERPRET GEOMETRIC TOLERANCING PER:			
MATERIAL			
FINISH			
NEXT ASSY	USED ON		
APPLICATION	DO NOT SCALE DRAWING		
		TITLE: Bowl	
		SIZE DWG. NO.	REV
		B	
		SCALE: 1:2 WEIGHT:	SHEET 1 OF 1

Project 3 | Technical drawing of Slide Pillar



	Details
CAD Software	Solidworks
Number of Components	1
Timeline	September 2026

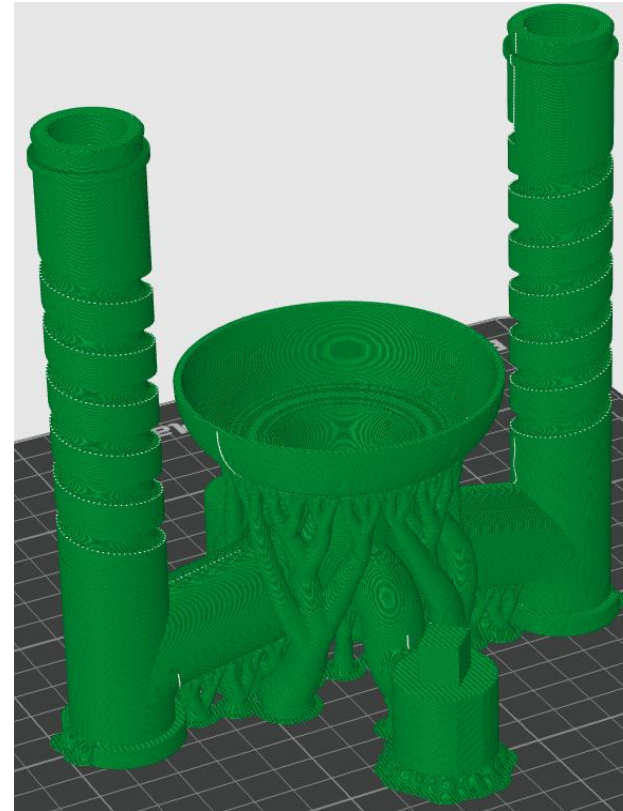
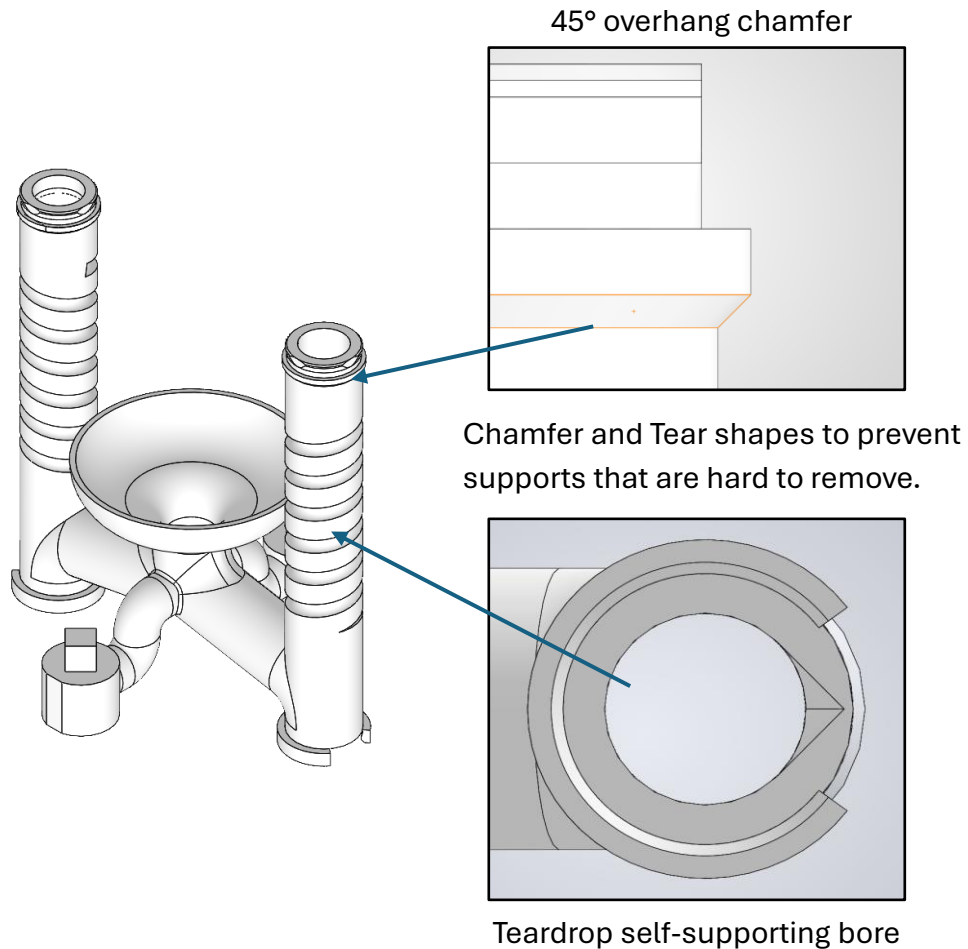
Project 3 | Technical drawing of Bucket



	Details
CAD Software	Solidworks
Number of Components	1
Timeline	September 2026

		UNLESS OTHERWISE SPECIFIED:	DRAWN	NAME	DATE	TITLE: Bucket		
		DIMENSIONS ARE IN MILLIMETERS	CHECKED	RYAN.H				
		TOLERANCES:	ENG APPR.					
		FRACTIONAL: $\pm$	MFG APPR.					
		ANGULAR: MACH: $\pm$ BEND: $\pm$	Q.A.			SIZE DWG. NO. REV		
		TWO PLACE DECIMAL: $\pm$	COMMENTS:					
		THREE PLACE DECIMAL: $\pm$				SCALE: 2:1 WEIGHT: SHEET 1 OF 1		
		INTERPRET GEOMETRIC TOLERANCING PER:						
		MATERIAL						
		FINISH						
NEXT ASSY	USED ON							
APPLICATION		DO NOT SCALE DRAWING						

Project 3 | Design Considerations for Additive Manufacturing



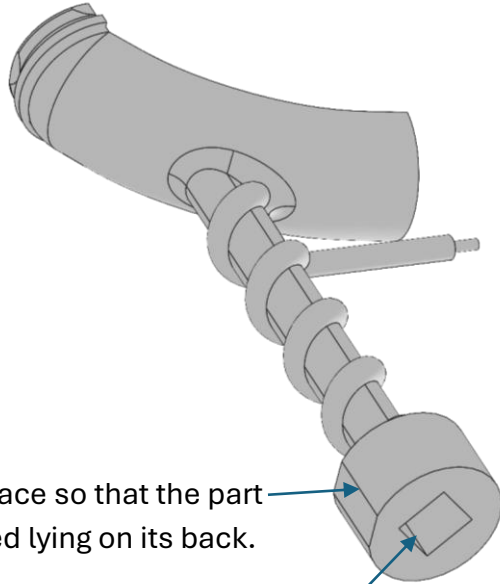
With Chamfer
Total Printing Time: 5h37m



Without Chamfer
Total Printing Time: 6h28m

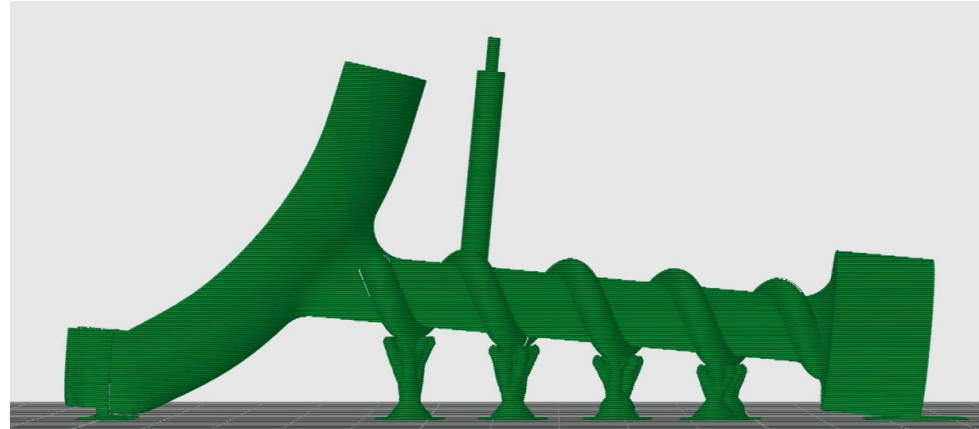
Integrated 45-degree chamfers and self-supporting teardrop profiles, eliminating internal support structures and cutting print cycle time by 51 minutes (13.1% reduction).

Project 3 | Design Considerations for Additive Manufacturing



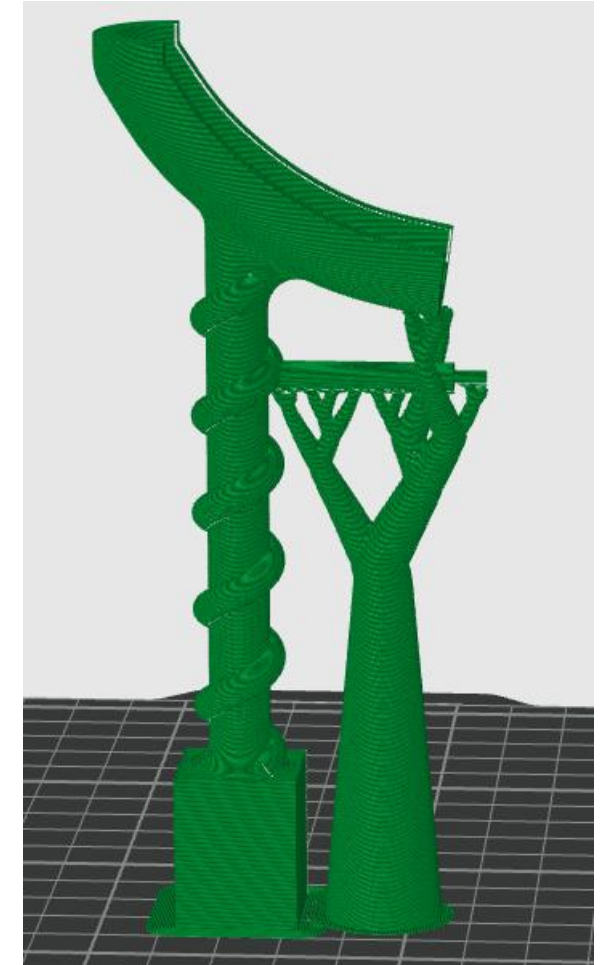
Added flat surface so that the part could be printed lying on its back.

Diamond shaped holes to reduce support.



Optimized Horizontal Build
Total Printing Time: 1h5m
Total Material Used: 19.64g

Just by adding a flat surface so that it could be printed at optimal orientation reduced the total printing time almost by **20mins** and material used by **7g**.



Unoptimized Vertical Build
(Extensive tree supports)
Total Printing Time: 1h24m
Total Material Used: 26.02g